Draft version February 19, 2026
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Homework[]:

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1. LINKS TO CODE

Here is a link to the code file https://github.com/HaikunXue/OU-ASTR-5900/blob/main/HW2/HW02.ipynb

2. QUESTION 1:

We have the equation:

$$x^3 - 7x^2 + 14x - 5 = 0$$

2.1. *Part a:*

Bisection method with initial guess $x_0 = 1$ and $x_1 = 1$:

Table 1. Bisection Method

x_{left}	x_{right}	x_{new}	y_{new}	x_{err}
0	1	0.5	0.375	0.5
0	0.5	0.25	-1.9219	0.25
0.25	0.5	0.375	-0.6816	0.125
0.375	0.5	0.4375	-0.1311	0.0625
0.4375	0.5	0.4688	0.1274	0.03125
0.4375	0.4688	0.4531	-4.692×10^{-4}	0.015625
0.4531	0.4688	0.4610	0.06381	0.0078125

And the result is shown in Table 1. We see a clear trend of y_{new} approaching zero, and a very close value at x = 0.4531.

2.2. *Part b:*

Newton - Raphson method with initial guess $x_0 = 0$. We use the relationship:

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

And the result is shown in Table 2. It is a much faster convergent!

2.3. *Part c:*

For Bisection method, it took 8 iterations. For Newton-Raphson Method, it took 3 iterations. Keep in mind though this is based upon a preset initial value. If we use different initial values the results might change dramatically, which we will find out in question 2.

Table 2. Newton-Raphson Method

x_i	x_{i+1}	y_{i+1}	x_{err}
0	0.3571	-0.8473	0.3571
0.3571	0.4474	-0.04761	0.09034
0.4474	0.4532	-1.880×10^{-4}	0.005759

	x_left	x_right	x_new	y_new	err
18	0.453178	0.453182	0.453180	-1.166298e-05	4.208807e-06
19	0.453180	0.453182	0.453181	-3.774580e-06	2.104399e-06
20	0.453181	0.453182	0.453182	1.696162e-07	1.052199e-06
21	0.453181	0.453182	0.453182	-1.802482e-06	5.260995e-07
22	0.453182	0.453182	0.453182	-8.164327e-07	2.630497e-07
23	0.453182	0.453182	0.453182	-3.234082e-07	1.315248e-07
24	0.453182	0.453182	0.453182	-7.689601e-08	6.576241e-08
25	0.453182	0.453182	0.453182	4.636010e-08	3.288120e-08
26	0.453182	0.453182	0.453182	-1.526795e-08	1.644060e-08
27	0.453182	0.453182	0.453182	1.554607e-08	8.220301e-09

Figure 1. Output of bisection method with initial value (0,1), last 10 rows out of 27 iterations.

3. QUESTION 2

In this section we will code both methods by hand, and approach the same problem.

3.1. *Part a:*

With the same initial conditions, for bisection method, it took me 27 iterations to reach the tolerance $\epsilon < 10^8$, the last 10 rows are shown in Figure 1. In comparison, the Newton Ralphson method took 5 iterations, as shown in Figure 2. The last y value reaches the display limit of 0.00! Impressive!

3.2. *Part b:*

Using integers from -5 to 6 as initial conditions, we ran Newton Raphson method and got the results shown

	x_ini	x_new	y_new	err
0	0.000000	0.357143	-8.473032e-01	1.000000e+00
1	0.357143	0.447448	-4.761133e-02	2.018229e-01
2	0.447448	0.453159	-1.843597e-04	1.260326e-02
3	0.453159	0.453182	-2.801860e-09	4.918040e-05
4	0.453182	0.453182	0.000000e+00	7.474562e-10

Figure 2. Output of Newton Raphson method with initial value 0. Five iterations and reaches the display limit!

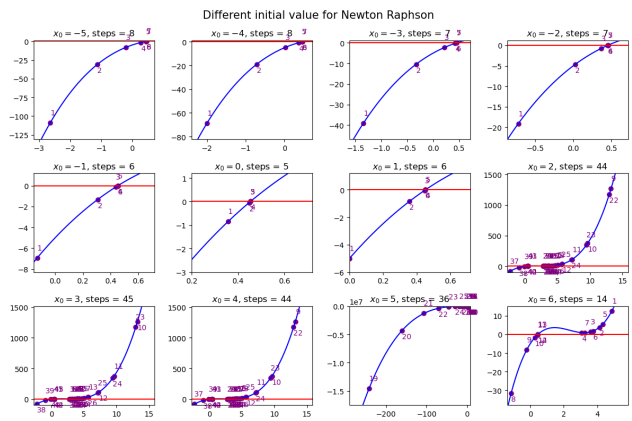

Figure 3. Output of Newton Raphson method with different integer initial values from -5 to 6.

in Figure 3. For values less than our root at 2, most method converges at less than 10 steps. However, for values greater than 2, some initial values lead to trapping within the local minimum at around x= 4, for x_0 = 5, we even overshoot to x = -200, then slowing return to root position. So the initial value does matter a lot for Newton Raphson method, and local minimum/maximum can cause issues. In contrast, bisection method has a more stable convergence.

3.3. *Part c:*

With more experiments, we took 1000 points between -5 and 6, and plotted the step vs initial value, as shown in Figure **??**. Similar to our result from Part b, it is well behaved below the root value at 0.45, and then rises exponentially around initial value of 2, taking as much as 200 iterations.

4. QUESTION 3

For the equation of interest, we are using Kepler's Third Law, this is interesting as it describes orbital motion of planets that is not perfectly circular but an ellipsoidal path. (in hindsight, might be too easy of an equation)

$$a^3 \propto T^2$$

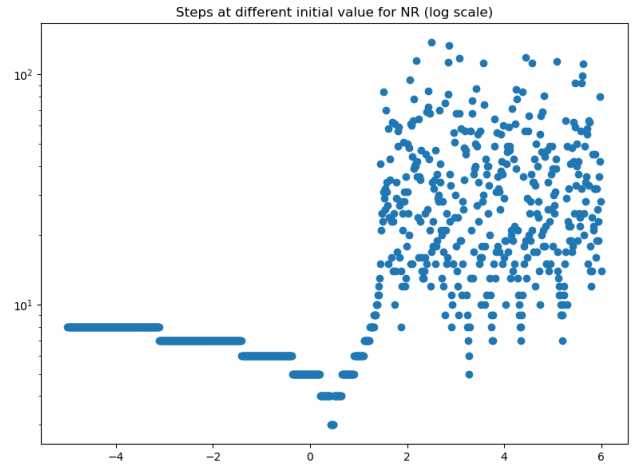

Figure 4. Newton Raphson method with 1000 different initial values between (-5,6), y is log scale.

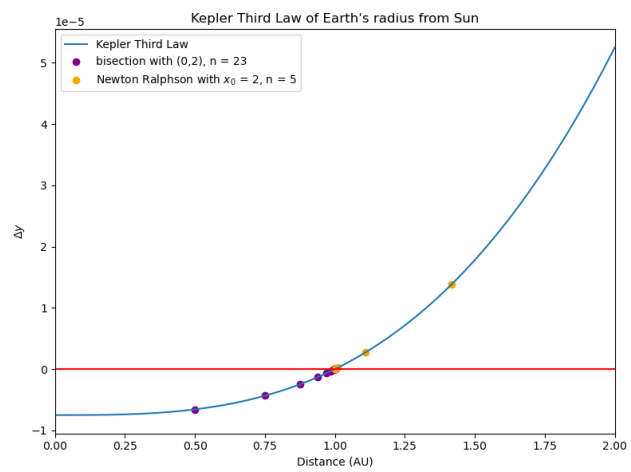

Figure 5. Kepler third law fit for Earth's orbit's radius. Newton Raphson is more efficient in this case.

relating the orbital period to the radius of planets. In equation form, we used

$$0 = \frac{a^3}{T^2} - 7.4958 \times 10^{-6}$$

where T is in days and a is in AU. With selected value of T = 365.25, we fitted a values with both methods and results are the following: Bisection took 23 iterations and Newton Raphson took 5 iteration, as expected for a simple power law function. The result is shown in Figure 5.

ACKNOWLEDGMENTS

Thanks to Professor Sean Matt for the template.